

# Draft Laporan UTS PostgreSQL

## Cover

**Judul:** Laporan UTS Basis Data – Analisis PostgreSQL

**Nama:** Raihan Muhammad Riswandi

**NIM:** 24110500001

**Mata Kuliah:** Advanced Basis Data

**Dosen Pengampu:** Hedy Pamungkas, S.T., M.T.I

**Tanggal:** 01/10/2025

---

## 1. Pendahuluan

Database yang diberikan dosen berisi informasi mengenai mahasiswa, dosen, mata kuliah, serta beasiswa. Laporan ini bertujuan untuk melakukan eksplorasi data menggunakan PostgreSQL dan menjawab pertanyaan analitis yang umum pada sistem akademik, yaitu:

1. Mahasiswa dengan GPA tertinggi di jurusan Computer Science (CS).
  2. Dosen yang mengajar paling banyak kelas.
  3. Total beasiswa yang dikeluarkan per semester.
  4. Daftar mahasiswa yang menerima beasiswa sekaligus financial aid.
- 

## 2. Analisis & Hasil Query

### 2.1 Mahasiswa dengan GPA Tertinggi di CS

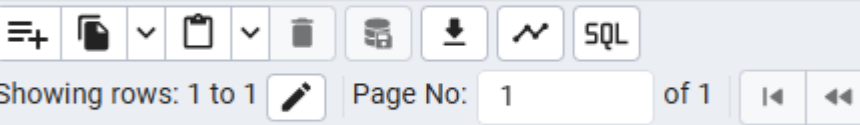
**Query SQL:**

```
SELECT s.id, s.name, s.gpa
FROM students s
JOIN departments d ON s.department_id = d.id
WHERE d.name = 'Computer Science'
ORDER BY s.gpa DESC
```

```
LIMIT 1;
```

## Hasil Analisis

```
1 SELECT s.id, s.name, s.gpa
2 FROM students s
3 JOIN departments d ON s.department_id = d.id
4 WHERE d.name = 'Computer Science'
5 ORDER BY s.gpa DESC
6 LIMIT 1;
7
```

| Data Output Messages Notifications                                                  |                    |                                 |                      |
|-------------------------------------------------------------------------------------|--------------------|---------------------------------|----------------------|
|  |                    |                                 |                      |
| Showing rows: 1 to 1 Page No: 1 of 1                                                |                    |                                 |                      |
|                                                                                     | id<br>[PK] integer | name<br>character varying (100) | gpa<br>numeric (3,2) |
| 1                                                                                   | 6                  | Lisa Chen                       | 3.92                 |

Mahasiswa dengan GPA tertinggi di jurusan CS adalah Lisa Chen dengan nilai GPA 3.92

---

## 2.2 Dosen dengan Jumlah Kelas Terbanyak

### Query SQL:

```
SELECT p.id, p.name, COUNT(ca.id) AS total_classes
FROM professors p
JOIN course_assignments ca ON p.id = ca.professor_id
GROUP BY p.id, p.name
ORDER BY total_classes DESC
LIMIT 1;
```

### Hasil Analisis:

```
1 SELECT p.id, p.name, COUNT(ca.id) AS total_classes
2 FROM professors p
3 JOIN course_assignments ca ON p.id = ca.professor_id
4 GROUP BY p.id, p.name
5 ORDER BY total_classes DESC
6 LIMIT 1;
7
```

|   | id<br>[PK] integer | name<br>character varying (100) | total_classes<br>bigint |
|---|--------------------|---------------------------------|-------------------------|
| 1 | 6                  | Dr. Lisa Wong                   | 3                       |

Dosen yang mengajar paling banyak kelas adalah Dr.Lisa Wong dengan jumlah kelas 3

### 2.3 Total Beasiswa yang Dikeluarkan per Semester

### Query SQL:

```
SELECT semester, SUM(amount) AS total_scholarships
FROM scholarships
GROUP BY semester;
```

### Hasil Analisis :

```
1 SELECT semester, SUM(amount) AS total_scholarships
2 FROM scholarships
3 GROUP BY semester;
4
```

| Data Output Messages Notifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                      |                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <div></div> |                                                                                                                      |                                                                                                                 |
| Showing rows: 1 to 1  Page No: 1 of 1                                                                                                                                                                                                                                                                                                    |                                                                                                                      |                                                                                                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | semester<br>character varying (10)  | total_scholarships<br>numeric  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2024-1                                                                                                               | 14000000.00                                                                                                     |

Terlihat bahwa beasiswa paling banyak dikeluarkan pada semester 1 dengan total 14.000.000

---

## 2.4 Mahasiswa yang Mendapat Beasiswa dan Financial Aid

### Query SQL:

```
SELECT DISTINCT s.id, s.name
FROM students s
JOIN scholarships sc ON s.id = sc.student_id
JOIN financial_aid fa ON s.id = fa.student_id;
```

### Hasil Analisis:

```
1 SELECT DISTINCT s.id, s.name
2 FROM students s
3 JOIN scholarships sc ON s.id = sc.student_id
4 JOIN financial_aid fa ON s.id = fa.student_id;
5
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

|   | Id<br>[PK] integer | name<br>character varying (100) |
|---|--------------------|---------------------------------|
| 1 | 2                  | Siti Rahayu                     |
| 2 | 1                  | Ahmad Sutrisno                  |
| 3 | 8                  | Sarah Johnson                   |
| 4 | 3                  | Budi Prakoso                    |

Jumlah mahasiswa yang mendapatkan keduanya adalah "Siti Rahayu", "Ahmad Sutrisno", "Sarah Johnson", "Budi Prakoso".

### 3. Diskusi

- Pertanyaan di atas memberikan gambaran mengenai distribusi prestasi mahasiswa, beban mengajar dosen, serta distribusi beasiswa.
- Analisis ini membantu melihat bagaimana sumber daya dialokasikan di universitas.

#### 4. Kesimpulan

1. Mahasiswa dengan GPA tertinggi di CS adalah **Lisa Chen(3.92)**
  2. Dosen dengan jumlah kelas terbanyak adalah **Dr. Lisa Wong**
  3. Total beasiswa tiap semester bervariasi, dengan jumlah terbesar di semester 1 (**14.000.000.**)
  4. Terdapat **4** mahasiswa yang menerima beasiswa sekaligus financial aid.
- 

#### 5. Lampiran

```
-- =====  
  
-- SECTION 1: DATABASE SCHEMA  
  
-- =====  
  
-- Drop existing tables if they exist  
  
DROP TABLE IF EXISTS course_assignments CASCADE;  
  
DROP TABLE IF EXISTS financial_aid CASCADE;  
  
DROP TABLE IF EXISTS scholarships CASCADE;  
  
DROP TABLE IF EXISTS enrollments CASCADE;  
  
DROP TABLE IF EXISTS courses CASCADE;  
  
DROP TABLE IF EXISTS students CASCADE;  
  
DROP TABLE IF EXISTS professors CASCADE;  
  
DROP TABLE IF EXISTS departments CASCADE;  
  
-- Create schema  
  
CREATE TABLE departments (
```

```
    id SERIAL PRIMARY KEY,

    name VARCHAR(100) NOT NULL,

    head_professor_id INTEGER,

    budget NUMERIC(12,2),

    established_year INTEGER

);

CREATE TABLE students (

    id SERIAL PRIMARY KEY,

    name VARCHAR(100) NOT NULL,

    email VARCHAR(100) UNIQUE,

    birth_date DATE,

    gpa NUMERIC(3,2),

    entry_semester VARCHAR(10),

    status VARCHAR(20) DEFAULT 'ACTIVE',

    department_id INTEGER REFERENCES departments(id)

);

CREATE TABLE professors (

    id SERIAL PRIMARY KEY,

    name VARCHAR(100) NOT NULL,

    department_id INTEGER REFERENCES departments(id),

    salary NUMERIC(10,2),

    hire_date DATE,

    specialization TEXT
```

```
);
```

```
CREATE TABLE courses (
```

```
    id SERIAL PRIMARY KEY,
```

```
    code VARCHAR(10) UNIQUE NOT NULL,
```

```
    name VARCHAR(100) NOT NULL,
```

```
    credits INTEGER CHECK (credits BETWEEN 1 AND 6),
```

```
    department_id INTEGER REFERENCES departments(id),
```

```
    level INTEGER CHECK (level IN (100, 200, 300, 400)),
```

```
    prerequisite_id INTEGER REFERENCES courses(id)
```

```
);
```

```
CREATE TABLE enrollments (
```

```
    id SERIAL PRIMARY KEY,
```

```
    student_id INTEGER REFERENCES students(id),
```

```
    course_id INTEGER REFERENCES courses(id),
```

```
    semester VARCHAR(10),
```

```
    grade CHAR(1) CHECK (grade IN ('A', 'B', 'C', 'D', 'F')),
```

```
    numeric_score NUMERIC(5,2),
```

```
    status VARCHAR(20) DEFAULT 'ENROLLED'
```

```
);
```

```
CREATE TABLE course_assignments (
```

```
    id SERIAL PRIMARY KEY,
```

```
    course_id INTEGER REFERENCES courses(id),
```



```

        professor_id INTEGER REFERENCES professors(id),

        semester VARCHAR(10),

        room VARCHAR(20),

        max_students INTEGER
    );

CREATE TABLE scholarships (

    id SERIAL PRIMARY KEY,

    student_id INTEGER REFERENCES students(id),

    type VARCHAR(50),

    amount NUMERIC(10,2),

    semester VARCHAR(10),

    criteria_met TEXT
);

CREATE TABLE financial_aid (

    id SERIAL PRIMARY KEY,

    student_id INTEGER REFERENCES students(id),

    family_income NUMERIC(12,2),

    aid_amount NUMERIC(10,2),

    semester VARCHAR(10),

    need_score INTEGER CHECK (need_score BETWEEN 0 AND 100)
);

```

```
-- =====
```

```
-- SECTION 2: SAMPLE DATA

-- =====

-- Insert departments

INSERT INTO departments (id, name, budget, established_year) VALUES

(1, 'Computer Science', 15000000.00, 1985),

(2, 'Mathematics', 12000000.00, 1980),

(3, 'Physics', 18000000.00, 1975),

(4, 'Business', 10000000.00, 1990);

-- Insert professors

INSERT INTO professors (id, name, department_id, salary, hire_date,
specialization) VALUES

(1, 'Dr. Ahmad Rahman', 1, 8500000, '2010-01-15', 'Database Systems'),

(2, 'Dr. Siti Nurhaliza', 1, 9200000, '2008-03-20', 'Machine
Learning'),

(3, 'Prof. Budi Santoso', 2, 10500000, '2005-08-10', 'Statistics'),

(4, 'Dr. Maria Santos', 3, 7800000, '2015-06-01', 'Quantum Physics'),

(5, 'Dr. John Smith', 4, 6500000, '2018-09-15', 'Marketing'),

(6, 'Dr. Lisa Wong', 1, 4500000, '2020-01-10', 'Software Engineering');

-- Insert courses

INSERT INTO courses (id, code, name, credits, department_id, level,
prerequisite_id) VALUES

(1, 'CS101', 'Introduction to Programming', 3, 1, 100, NULL),

(2, 'CS201', 'Data Structures', 3, 1, 200, 1),
```

```
(3, 'CS301', 'Database Systems', 3, 1, 300, 2),
(4, 'CS401', 'Advanced Database', 3, 1, 400, 3),
(5, 'MATH101', 'Calculus I', 4, 2, 100, NULL),
(6, 'MATH201', 'Statistics', 3, 2, 200, 5),
(7, 'PHY101', 'Physics I', 4, 3, 100, NULL),
(8, 'BUS101', 'Business Fundamentals', 2, 4, 100, NULL),
(9, 'CS102', 'Programming II', 3, 1, 100, 1),
(10, 'CS302', 'Software Engineering', 3, 1, 300, 2);

-- Insert students

INSERT INTO students (id, name, email, birth_date, gpa, entry_semester,
status, department_id) VALUES

(1, 'Ahmad Sutrisno', 'ahmad@student.edu', '2002-05-15', 3.85,
'2021-1', 'ACTIVE', 1),

(2, 'Siti Rahayu', 'siti@student.edu', '2001-08-20', 3.45, '2020-2',
'ACTIVE', 1),

(3, 'Budi Prakoso', 'budi@student.edu', '2002-12-10', 2.95, '2021-1',
'ACTIVE', 2),

(4, 'Maria Gonzalez', 'maria@student.edu', '2001-03-25', 3.75,
'2020-1', 'ACTIVE', 3),

(5, 'John Doe', 'john@student.edu', '2003-01-08', 2.15, '2022-1',
'ACTIVE', 4),

(6, 'Lisa Chen', 'lisa@student.edu', '2002-09-18', 3.92, '2021-2',
'ACTIVE', 1),

(7, 'Robert Wilson', 'robert@student.edu', '2001-11-30', 1.85,
'2020-1', 'ACTIVE', 2),

(8, 'Sarah Johnson', 'sarah@student.edu', '2002-07-12', 3.25, '2021-2',
'ACTIVE', 3);
```

```
-- Insert enrollments

INSERT INTO enrollments (id, student_id, course_id, semester, grade,
numeric_score, status) VALUES

-- Ahmad Sutrisno (student 1) - excellent student

(1, 1, 1, '2021-1', 'A', 92.5, 'COMPLETED'),
(2, 1, 2, '2021-2', 'A', 88.0, 'COMPLETED'),
(3, 1, 3, '2022-1', 'B', 85.5, 'COMPLETED'),
(4, 1, 4, '2022-2', 'A', 90.0, 'COMPLETED'),
(5, 1, 5, '2021-1', 'B', 82.0, 'COMPLETED'),

-- Siti Rahayu (student 2) - good student

(6, 2, 1, '2020-2', 'B', 78.5, 'COMPLETED'),
(7, 2, 2, '2021-1', 'B', 81.0, 'COMPLETED'),
(8, 2, 3, '2021-2', 'C', 72.5, 'COMPLETED'),
(9, 2, 5, '2020-2', 'A', 89.0, 'COMPLETED'),

-- Budi Prakoso (student 3) - average student

(10, 3, 5, '2021-1', 'C', 65.0, 'COMPLETED'),
(11, 3, 6, '2021-2', 'D', 58.5, 'COMPLETED'),
(12, 3, 1, '2022-1', 'B', 75.0, 'COMPLETED'),

-- Maria Gonzalez (student 4) - good student

(13, 4, 7, '2020-1', 'A', 91.5, 'COMPLETED'),
(14, 4, 5, '2020-2', 'B', 83.0, 'COMPLETED'),
(15, 4, 1, '2021-1', 'A', 87.5, 'COMPLETED'),
```

```
-- John Doe (student 5) - poor student

(16, 5, 8, '2022-1', 'D', 55.0, 'COMPLETED'),
(17, 5, 1, '2022-2', 'F', 42.0, 'COMPLETED'),
(18, 5, 5, '2023-1', 'D', 59.5, 'COMPLETED'),

-- Lisa Chen (student 6) - excellent student

(19, 6, 1, '2021-2', 'A', 95.0, 'COMPLETED'),
(20, 6, 2, '2022-1', 'A', 93.5, 'COMPLETED'),
(21, 6, 3, '2022-2', 'A', 91.0, 'COMPLETED'),
(22, 6, 10, '2023-1', 'A', 89.5, 'COMPLETED'),

-- Robert Wilson (student 7) - poor student

(23, 7, 5, '2020-1', 'F', 35.0, 'COMPLETED'),
(24, 7, 1, '2020-2', 'D', 57.5, 'COMPLETED'),
(25, 7, 6, '2021-1', 'F', 41.0, 'COMPLETED'),
(26, 7, 1, '2021-2', 'F', 45.0, 'COMPLETED'), -- Retake

-- Sarah Johnson (student 8) - average student

(27, 8, 7, '2021-2', 'B', 76.0, 'COMPLETED'),
(28, 8, 5, '2022-1', 'C', 68.5, 'COMPLETED'),
(29, 8, 1, '2022-2', 'B', 79.0, 'COMPLETED');

-- Insert course assignments
```

```

INSERT INTO course_assignments (id, course_id, professor_id, semester,
room, max_students) VALUES

(1, 1, 1, '2024-1', 'LAB-101', 30),

(2, 1, 6, '2024-1', 'LAB-102', 25),  -- CS101 has 2 classes

(3, 2, 1, '2024-1', 'LAB-201', 35),

(4, 3, 1, '2024-1', 'LAB-301', 40),

(5, 4, 2, '2024-1', 'LAB-401', 20),

(6, 5, 3, '2024-1', 'MATH-101', 50),

(7, 6, 3, '2024-1', 'MATH-201', 30),

(8, 7, 4, '2024-1', 'PHY-101', 45),

(9, 8, 5, '2024-1', 'BUS-101', 40),

(10, 9, 6, '2024-1', 'LAB-103', 25),

(11, 10, 6, '2024-1', 'LAB-302', 30),

(12, 10, 2, '2024-1', 'LAB-303', 25);  -- CS302 has 2 classes


-- Insert scholarships

INSERT INTO scholarships (id, student_id, type, amount, semester,
criteria_met) VALUES

(1, 1, 'MERIT_HIGH', 3000000, '2024-1', 'GPA >= 3.7'),

(2, 4, 'MERIT_HIGH', 3000000, '2024-1', 'GPA >= 3.7'),

(3, 6, 'MERIT_HIGH', 3000000, '2024-1', 'GPA >= 3.7'),

(4, 2, 'MERIT_MEDIUM', 2000000, '2024-1', 'GPA 3.3-3.69'),

(5, 8, 'MERIT_MEDIUM', 2000000, '2024-1', 'GPA 3.3-3.69'),

(6, 3, 'MERIT_LOW', 1000000, '2024-1', 'GPA 3.0-3.29');


-- Insert financial aid

```

```
INSERT INTO financial_aid (id, student_id, family_income, aid_amount,
semester, need_score) VALUES

(1, 1, 25000000, 1500000, '2024-1', 75),

(2, 2, 18000000, 2000000, '2024-1', 85),

(3, 3, 15000000, 2500000, '2024-1', 90),

(4, 5, 12000000, 3000000, '2024-1', 95),

(5, 7, 10000000, 3500000, '2024-1', 98),

(6, 8, 22000000, 1800000, '2024-1', 80);
```